Type MS · Integrating trig functions using identities

Answer 1 of 9 · Question 6 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

number		
6 (a)	$(\sin A\cos B+\cos A\sin B)+(\sin A\cos B-\cos A\sin B)$ $=2\sin A\cos B$ *	M1 A1 cso (2)
(b)	$\sin 8x+\sin 6x$	B1 (1)
(c)	$3\int_0^{\frac{\pi}{4}}(\sin 8x+\sin 6x)\,dx$ $=(3)\left[-\frac{1}{8}\cos 8x-\frac{1}{6}\cos 6x\right]_0^{\frac{\pi}{4}}$ $=(3)\left[\left(-\frac{1}{8}-0\right)-\left(-\frac{1}{8}-\frac{1}{6}\right)\right]$ $=\frac{1}{2}$ cao oe	M1 A1 M1 A1cao (4) [7]

Part	Mark	Additional Guidance
(a)	M1	Correct expression show.
	A1	cso
(b)	B1	For the expression shown
(c)	M1	$k\int_0^{\frac{\pi}{4}}(\sin 8x+\sin 6x)\,dx$ $k\neq 0$ or 1 k must be an integer. This mark can be awarded if the limits aren't seen on the integral.
	A1	Correctly integrated, the 3 and the limits need not be present for this mark
	M1	Correctly shown substitution of limits, with a subtract sign between. The 3 need not be present for this mark, need not be simplified. *This mark can be implied if first M1 A1 awarded and final correct answer.* Allow a correct substitution of limits into any changed expression.
	A1	cao oe

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Answer 1 of 9 · Question 6 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Integrating trig functions using identities

Answer 2 of 9 · Question 1 part whole · 4 marks

2022 JAN · Paper 01 · extract 1 of 7 · find part whole in this mark scheme extract

Paper 1

Type MS · Integrating trig functions using identities

Answer 2 of 9 · Question 1 part whole · 4 marks

2022 JAN · Paper 01 · extract 2 of 7 · find part whole in this mark scheme extract

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Type MS · Integrating trig functions using identities

Answer 2 of 9 · Question 1 part whole · 4 marks

2022 JAN · Paper 01 · extract 3 of 7 · find part whole in this mark scheme extract

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the last candidate in exactly the same way as they mark the first.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme - not according to their perception of where the grade boundaries may lie.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification/indicative content will not be exhaustive.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, a senior examiner must be consulted before a mark is given.
- Crossed out work should be marked **unless** the candidate has replaced it with an alternative response.

- **Types of mark**
  - M marks: method marks
  - A marks: accuracy marks
  - B marks: unconditional accuracy marks (independent of M marks)

- **Abbreviations**
  - cao – correct answer only
  - ft – follow through
  - isw – ignore subsequent working
  - SC - special case
  - oe – or equivalent (and appropriate)
  - dep – dependent
  - indep – independent
  - awrt – answer which rounds to
  - eeoo – each error or omission

- **No working**
  If no working is shown then correct answers normally score full marks
  If no working is shown then incorrect (even though nearly correct) answers score no marks.

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

- **With working**
  You must always check the working in the body of the script (and on any diagrams) irrespective of whether the final answer is correct or incorrect and award any marks appropriate from the mark scheme.
  If it is clear from the working that the "correct" answer has been obtained from incorrect working, award 0 marks.
  If a candidate misreads a number from the question. Eg. Uses 252 instead of 255; method marks may be awarded provided the question has not been simplified. Examiners should send any instance of a suspected misread to review.
  If there is a choice of methods shown, then award the lowest mark, unless the answer on the answer line makes clear the method that has been used.
  If there is no answer achieved then check the working for any marks appropriate from the mark scheme.

- **Ignoring subsequent work**
  It is appropriate to ignore subsequent work when the additional work does not change the answer in a way that is inappropriate for the question: eg. Incorrect cancelling of a fraction that would otherwise be correct.
  It is not appropriate to ignore subsequent work when the additional work essentially makes the answer incorrect eg algebra.
  Transcription errors occur when candidates present a correct answer in working, and write it incorrectly on the answer line; mark the correct answer.

- **Parts of questions**
  Unless allowed by the mark scheme, the marks allocated to one part of the question CANNOT be awarded to another.

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Type MS · Integrating trig functions using identities

Answer 2 of 9 · Question 1 part whole · 4 marks

2022 JAN · Paper 01 · extract 5 of 7 · find part whole in this mark scheme extract

General Principles for Further Pure Mathematics Marking

(but note that specific mark schemes may sometimes override these general principles)

Method mark for solving a 3 term quadratic equation:

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0, \quad q\neq 0$ leading to $x=....$

4. Use of calculators

Unless the question specifically states 'show' or 'prove' accept correct answers from no working. If an incorrect solution is given without any working do not award the Method mark.

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

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General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show....")

Exact answers:
When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)
Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

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Type MS · Integrating trig functions using identities

Answer 2 of 9 · Question 1 part whole · 4 marks

2022 JAN · Paper 01 · extract 7 of 7 · find part whole in this mark scheme extract

edexcel

January 2022
4PM1 Paper 1
Mark Scheme

Question	Scheme	Marks
1	$\int \cos 4\theta \ d\theta = \left[\frac{\sin 4\theta}{4}\right]$ For an attempt to evaluate **their** integral using the given values and reach a value	M1A1
	$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos 4\theta \ d\theta = \left[\frac{\sin 4\theta}{4}\right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} = \frac{\sin\left(4\times\frac{\pi}{3}\right)}{4} - \frac{\sin\left(4\times\frac{\pi}{4}\right)}{4} = \ldots$	M1
	$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cos 4\theta \ d\theta = -\frac{\sqrt{3}}{8}$	A1 [4]
		Total 4 marks

Type MS · Integrating trig functions using identities

Answer 3 of 9 · Question 10 part d · 4 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 4 · find part d in this mark scheme extract

Number		
10 (a)	$\cos 2A = \cos^2 A - \sin^2 A = \cos^2 A - \left(1 - \cos^2 A\right) = 2\cos^2 A - 1*$	M1 A1cso [2]
(b)	$\cos 4A = 2\cos^2 2A - 1$ $= 2\left(2\cos^2 A - 1\right)^2 - 1$ $= 2\left(4\cos^4 A - 4\cos^2 A + 1\right) - 1$ $= 8\cos^4 A - 8\cos^2 A + 1*$	M1 M1 M1 A1cso [4]
(c)	Let $\frac{\theta}{4} + \frac{\pi}{24} = x$ $\cos^2 x\left(\cos^2 x - 1\right) = \cos^4 x - \cos^2 x$ $\Rightarrow \cos^4 x - \cos^2 x = \frac{\cos 4x - 1}{8}$ $4\left(\frac{\theta}{4} + \frac{\pi}{24}\right) = \theta + \frac{\pi}{6}$ $\frac{\cos\left(\theta + \frac{\pi}{6}\right) - 1}{8} = -\frac{1}{16} \Rightarrow \cos\left(\theta + \frac{\pi}{6}\right) = \frac{1}{2}$ $\Rightarrow \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3} \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1 M1 M1 A1 [5]
Alternative (with x defined as above)		
(c)	$\cos^4 x - \cos^2 x = -\frac{1}{16} \Rightarrow 8\cos^4 x - 8\cos^2 x + 1 = 1 - \frac{8}{16}$ $\cos 4x = \frac{1}{2} \Rightarrow 4x = \theta + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \cdots \Rightarrow \theta = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1M1 M1A1
Alternative (with x defined as above)		
(c)	$16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0$ $\cos^2 x = \frac{-(-16) \pm \sqrt{(-16)^2 - 4\times16\times1}}{2\times16} = \frac{2 \pm \sqrt{3}}{4}$ $\cos x = \sqrt{\frac{2 \pm \sqrt{3}}{4}}$ $(= \pm 0.9659...$ or $\pm 0.2588...)$ $x = 0.2617..., 1.3089..., 1.8325..., 2.8787...\left(= \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}...\right)$ $\theta = 4\left(x - \frac{\pi}{24}\right) = \frac{\pi}{6}, \frac{3\pi}{2}$	M1 M1 M1 M1A1
(d)	$\mathrm{f}(A) = \frac{\cos 4A}{2} + \frac{1}{2}$ $\left[\frac{\sin 4A}{8} + \frac{A}{2}\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \left(\frac{\sin 2\pi}{8} + \frac{\pi}{4}\right) - \left(\frac{\sin\frac{2\pi}{3}}{8} + \frac{\pi}{12}\right) = \frac{\pi}{4} - \frac{\sqrt{3}}{16} - \frac{\pi}{12} = \frac{\pi}{6} - \frac{\sqrt{3}}{16}$	M1 dM1 ddM1 A1 [4]
		Total 15 marks

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Answer 3 of 9 · Question 10 part d · 4 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 4 · find part d in this mark scheme extract

(a)	M1	Use $\cos(A+A)=\cos A\cos A-\sin A\sin A$ (may go straight to $\cos 2A=\cos^2 A-\sin^2 A$) and then try to replace $\sin^2 A$ by $1-\cos^2 A$ Allow with any other single variable.
	A1 cso	Obtain $\cos 2A=2\cos^2 A-1$ with no errors in the working. Must be stated with A as the variable.
(b)		Allow with any other single variable for all M marks. Allow one sign error removing brackets and ft for all M marks.
	M1	$\cos 4A=2\cos^2 2A-1$ or $\cos 4A=\cos^2 2A-\sin^2 2A$ Obtain an expression in terms of cos 2A and/or sin 2A.
	M1	eg $\cos 4A=2\left(2\cos^2 A-1\right)^2-1$, $\cos 4A=\left(2\cos^2 A-1\right)^2-4\left(1-\cos^2 A\right)\cos^2 A$ Obtain a correct expression in cos A. Eliminate terms in sin A and sin 2A.
	M1	Expand brackets. Numerical factors may remain. e.g. $\cos 4A=2\left(4\cos^4 A-2\cos^2 A-2\cos^2 A+1\right)-1$ e.g. $\cos 4A=4\cos^4 A-4\cos^2 A+1-4\cos^2 A+4\cos^4 A$
	A1 cso	Obtain the given result with no errors in the working. Must be stated with A as the variable.
(c)	M1	Expand brackets and use the result from part (b) to express the LHS of the equation in terms of $4x$, $\frac{\cos 4x-1}{8}$, where $x=\frac{\theta}{4}+\frac{\pi}{24}$ **Alternative** Arrange the equation in the form $8\cos^4 x-8\cos^2 x+1=1-\frac{8}{16}$ Allow a single numerical or sign error in the constants for both methods.
	M1	Rearrange the equation to give $\cos\left(\theta+\frac{\pi}{6}\right)=\frac{1}{2}$ or oe Note that this mark is shown as B1 on ePEN.
	M1	Find any correct positive value for $4\left(\frac{\theta}{4}+\frac{\pi}{24}\right)$, given in radians. eg $\frac{\pi}{3}$
	M1	Find any correct positive value for θ, given in radians. eg $\frac{\pi}{6}$
	A1	Both correct answers with no extras, given in radians as multiples of π. Ignore answers outside the given range.

Type MS · Integrating trig functions using identities

Answer 3 of 9 · Question 10 part d · 4 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 4 · find part d in this mark scheme extract

Alternative		
(c)	M1	Write the equation as a three term quadratic. e.g. $16\left(\cos^2 x\right)^2 - 16\cos^2 x + 1 = 0, \left(x = \frac{\theta}{4} + \frac{\pi}{24}\right)$, $A^2 - A + \frac{1}{16} = 0, \left(A = \cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\right)$
	M1	Any valid method to solve their quadratic equation. Note that this mark is shown as B1 on ePEN.
	M1	Any correct value for $x = \frac{\theta}{4} + \frac{\pi}{24}$ in the range $0 \le x \le \pi$, given in radians, either as a decimal or as a multiple of π.
	M1	Either value of θ given in radians, $\theta = \frac{\pi}{6}$ or $\theta = \frac{3\pi}{2}$ or $\theta = 0.523...$ or $\theta = 4.712...$
	A1	Both $\theta = \frac{\pi}{6}$ and $\theta = \frac{3\pi}{2}$ given as multiples of π.
(d)	M1	Use result from part (b) to write $\mathrm{f}(A) = k(\cos 4A \pm 1)$.
	dM1	Integrate their $\mathrm{f}(A)$ correctly, dep on M1 awarded. Ignore limits.
	ddM1	Substitute both given limits, dep on both M marks awarded.
	A1	$\frac{\pi}{6} - \frac{\sqrt{3}}{16}$ *or* $\frac{1}{6}\pi - \frac{1}{16}\sqrt{3}$ simplified in this format. Do not isw.

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Answer 3 of 9 · Question 10 part d · 4 marks

2019 JUN · Paper 02 (regional R) · extract 4 of 4 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Integrating trig functions using identities

Answer 4 of 9 · Question 10 part e · 4 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part e in this mark scheme extract

10(a)	$\cos(A+B)+\cos(A-B)=\cos A\cos B-\sin A\sin B+\cos A\cos B+\sin A\sin B$	M1
	$=2\cos A\cos B$ *	A1 cso (2)
(b)	Let $A+B=P,\quad A-B=Q \Rightarrow A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)$	M1
	As $\cos(A+B)+\cos(A-B)=2\cos A\cos B$	
	$\cos P+\cos Q=2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)$ *	M1A1cso (3)
ALT		
	Let $A=\frac{1}{2}(P+Q),\quad B=\frac{1}{2}(P-Q)\Rightarrow A+B=P,\ A-B=Q$	M1
	As $2\cos A\cos B=\cos(A+B)+\cos(A-B)$	
	$2\cos\frac{1}{2}(P+Q)\cos\frac{1}{2}(P-Q)=\cos P+\cos Q$ *	M1A1cso
(c)	$\cos 5\theta+\cos 7\theta=2\cos 6\theta\cos\theta=0$	M1
	$\cos 6\theta=0\Rightarrow 6\theta=\frac{\pi}{2},\frac{3\pi}{2},\frac{5\pi}{2}\Rightarrow\theta=\frac{\pi}{12},\frac{\pi}{4}\ \left(\text{or}\ \frac{3\pi}{12}\right),\frac{5\pi}{12}$	A1A1
	$\cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$	A1 (4)
(d)	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 6x)+(\cos 6x+\cos 4x)$	
	$=2\cos 7x\cos x+2\cos 5x\cos x$	M1
	$=2\cos x(\cos 7x+\cos 5x)=2\cos x\times 2(\cos 6x\cos x),\ =4\cos 6x\cos^2 x$ *	dM1,A1cso (3)
ALT 1	$\cos 8x+2\cos 6x+\cos 4x=(\cos 8x+\cos 4x)+2\cos 6x$	
	$=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=2\cos 6x(\cos 2x+1)=2\cos 6x(2\cos^2 x-1+1)=4\cos 6x\cos^2 x$ *	dM1A1cso (3)
ALT 2	Working from right to left	
	$4\cos 6x\cos^2 x=4\cos 6x\times\frac{1}{2}(\cos 2x+1)=2\cos 6x\cos 2x+2\cos 6x$	M1
	$=\cos 8x+\cos 4x+2\cos 6x$ *	dM1A1cso (3)

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Answer 4 of 9 · Question 10 part e · 4 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part e in this mark scheme extract

(e)	$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \frac{1}{4}\int_0^{\frac{\pi}{3}} (\cos 8x + 2\cos 6x + \cos 4x) \, dx$	M1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin 8x + \frac{1}{3}\sin 6x + \frac{1}{4}\sin 4x\right]_0^{\frac{\pi}{3}}$	A1	
	$= \frac{1}{4}\left[\frac{1}{8}\sin\frac{8\pi}{3} + \frac{1}{3}\sin 2\pi + \frac{1}{4}\sin\frac{4\pi}{3} - 0\right] = -\frac{\sqrt{3}}{64}$ (-0.02706... scores M1A0)	dM1A1	(4)
			[16]
ALT	For the first M mark Award M1 only when the integrand has been changed (by a valid method) to a function which can be integrated $\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx = \int_0^{\frac{\pi}{3}} \cos 6x \times \frac{1}{2}(\cos 2x + 1) \, dx$		
	$= \frac{1}{2}\int_0^{\frac{\pi}{3}} (\cos 6x \times \cos 2x + \cos 6x) \, dx = \frac{1}{2}\int_0^{\frac{\pi}{3}} \frac{1}{2}(\cos 8x + \cos 4x + 2\cos 6x) \, dx$	M1	
	Rest as above.		

(a)	
M1	Use the standard formulae. This is a "show that" question so these formulae must be written out in full. Both sides of the result must be seen although the working may appear between them as seen in the scheme.
A1cso	Final **given** result. Do not award if the expansions shown are not in the correct order (this suggests incorrect signs in the formulae).
(b)	
M1	Use $A+B=P,\quad A-B=Q$ to obtain A and B in terms of P and Q
M1	Substitute in the result from (a) to obtain an identity in P and Q only
A1cso	Correct result reached with no errors seen
ALT	Working right to left: Notes similar to above
(c)	
M1	Use the result from (b) to show that $(2)\cos 6\theta \cos(\pm\theta) = 0$
A1	Obtain one correct solution of $\cos 6\theta = 0$. Allow if in decimal form but must be radians.
A1	Two further correct solutions **and no more within the range.** If any of the 3 solutions of $\cos 6\theta = 0$ is not exact do not award this mark.
A1	State the solution of $\cos(\pm\theta) = 0$. Must be exact unless this is penalised above.
NB	If answers are given in degrees deduct 2A marks from any that would otherwise have been given. (Answers in degrees but then changed to radians are acceptable – mark the radian answers.) Ignore extra answers outside the stated range – any within are incorrect.
(d)	
M1	Use the result from (a) or (b) once. Allow if one or both "2"s are missing
dM1	Use the result from (a) or (b) again. Must include both "2"s this time. Depends on previous M mark.
A1cso	Reach the **given** result with no errors seen

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Answer 4 of 9 · Question 10 part e · 4 marks

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ALT 1	
M1	Use the result from (a) or (b). Allow if the "2" is missing.
dM1	Factorise and use the *correct* double angle formula on cos2x Depends on the previous M mark
A1cso	Reach the **given** result with no errors seen
ALT 2	
M1	Use the *correct* half angle formula on $\cos^2 x$
dM1A1	M1: Use the result from (a) or (b)
cso	A1: reach the **given** result with no errors seen
(e)M1	Obtain a function which can be integrated **either** by using the **given** result from (d) **OR**. deriving the same result. Allow if 1/4 is missing. (Integration by Parts – send to review)
A1	Correct integration (must have included the ¼) Limits not needed for these 2 marks.
dM1	Correct use of the given limits. All sines are 0 at the lower limit so these need not be shown
A1	Correct final answer which must be **exact and stated as a single fraction**.

Type MS · Integrating trig functions using identities

Answer 5 of 9 · Question 11 part c · 4 marks

2025 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

11 (a)	$\cos^4\theta-\sin^4\theta=(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$	M1
	$=(\cos^2\theta-\sin^2\theta)\times1=\cos2\theta$ *	M1M1A1 cso
	ALT example	[4
	$\cos^4\theta-\sin^4\theta=\cos^4\theta-\sin^2\theta\sin^2\theta$	
	$=\cos^4\theta-(1-\cos^2\theta)(1-\cos^2\theta)$	2nd M1
	$=\cos^4\theta-(1-2\cos^2\theta+\cos^4\theta)$	1st M1
	$=(1-2\cos^2\theta)$	3rd M1
	$=\cos2\theta$	A1
(b)	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-3=2[\cos^4\left(\theta+\frac{\pi}{8}\right)-\sin^4\left(\theta+\frac{\pi}{8}\right)]$	M1
	$=2\cos\left(2\theta+\frac{\pi}{4}\right)$	A1
	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-2\cos\left(2\theta+\frac{\pi}{4}\right)-3=0$	M1
	$\left(4\cos\left(2\theta+\frac{\pi}{4}\right)-3\right)\left(2\cos\left(2\theta+\frac{\pi}{4}\right)+1\right)[=0]$	
	$\cos\left(2\theta+\frac{\pi}{4}\right)=\frac{3}{4}\quad\cos\left(2\theta+\frac{\pi}{4}\right)=-\frac{1}{2}$	A1
	$2\theta+\frac{\pi}{4}=0.72273...\quad2\theta+\frac{\pi}{4}=\frac{2\pi}{3}$	dM1
	$\theta=2.39,3.11,\ 0.65,1.70$ accept $\theta=\frac{5\pi}{24},\frac{13\pi}{24}$	A1A1 [7]
(c)	$\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos^4 2x-\sin^4 2x-8\sin4x\right)\mathrm{d}x=\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos4x-8\sin4x\right)\mathrm{d}x$	M1
	$=\left[\frac{1}{4}\sin4x+\frac{8}{4}\cos4x\right]_{\frac{\pi}{16}}^{\frac{\pi}{8}}$	M1
	$=\frac{1}{4}\left[\sin\left(4\times\frac{\pi}{8}\right)-\sin\left(4\times\frac{\pi}{16}\right)\right]+2\left[\cos\left(4\times\frac{\pi}{8}\right)-\cos\left(4\times\frac{\pi}{16}\right)\right]$	M1
	$=\frac{1}{4}\left[1-\frac{\sqrt{2}}{2}\right]+2\left[0-\frac{\sqrt{2}}{2}\right]=\frac{1}{4}-\frac{9\sqrt{2}}{8}$	A1 [4]
		Total 15 marks

Type MS · Integrating trig functions using identities

Answer 5 of 9 · Question 11 part c · 4 marks

2025 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
General Principles for marking part (a) • First M mark is for either applying the difference of two squares or multiplying out their expression and collecting terms. • Second M mark is for applying $\cos^2\theta+\sin^2\theta=1$ in any form. This can be implied from correct work. However, do not award for just $\cos^4\theta-\sin^4\theta=\cos^2\theta-\sin^2\theta$ seen. • Third M mark is for applying any identity for $\cos 2\theta$ • A1 For obtaining the required result with no errors.		
(a)	M1	For factorising as the difference of two squares $(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$
	M1	For use of $\cos^2\theta+\sin^2\theta)=1$
	M1	For use of $\cos^2\theta-\sin^2\theta=\cos(2\theta)$
	A1 cso	Obtain the given result with no errors in the working.
(b)	M1	For using the given result to obtain a quadratic in $\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	A1	For rearranging to the correct quadratic
	M1	For a valid attempt to solve their **3TQ**. See General Guidance for what constitutes an attempt to factorise. If candidates use a calculator [so no working shown] then award this mark only if the 3TQ is correct with the correct solutions.
	A1	For obtaining correct values for $\cos\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	dM1	For achieving any valid value **in radians** for their $2\theta+\frac{\pi}{4}$ i.e. 0.7227…or $\frac{2}{3}\pi$ Accept values out of range for this mark. This is dependent on the previous M mark.
	A1	For two correct solutions in range
	A1	For all four correct solutions in range and no others. Extra solutions within range scores A0 Ignore solutions outside of range. Exact solutions when given must be given as multiples of π.
(c)	M1	For use of the given result to obtain $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\cos 4x\ \mathrm{d}x$
	M1	For an attempt to integrate $\cos kx\rightarrow l\sin kx$ where $k,l\neq 1$
	M1	For substituting limits into a changed expression
	A1	For the correct answer in the required form.

Type MS · Integrating trig functions using identities

Answer 5 of 9 · Question 11 part c · 4 marks

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Type MS · Integrating trig functions using identities

Answer 6 of 9 · Question 11 part d · 4 marks

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11(a)(i)	$\cos 2A = \cos^2 A - \sin^2 A, = \cos^2 A - \left(1 - \cos^2 A\right)$ $\cos 2A = 2\cos^2 A - 1$ *	M1,M1 A1 cso [3]
(ii)	$\sin 2A = \sin A\cos A + \sin A\cos A = 2\sin A\cos A$ *	B1 cso [1]
(b)	$\cos 3A = \cos\left(2A + A\right)$ $= \cos 2A\cos A - \sin A\sin 2A$ $= \left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A$ $= 2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A$ $= 2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A$ $= 4\cos^3 A - 3\cos A$ $\Rightarrow \cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ * **ALT** $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ $= \dfrac{\cos 2A\cos A - \sin 2A\sin A + 3\cos A}{4}$ $= \dfrac{\left(2\cos^2 A - 1\right)\cos A - 2\sin A\sin A\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A - 2\left(1 - \cos^2 A\right)\cos A + 3\cos A}{4}$ $= \dfrac{2\cos^3 A - \cos A + 2\cos^3 A - 2\cos A + 3\cos A}{4}$ $= \dfrac{4\cos^3 A}{4} = \cos^3 A \quad [LHS = RHS]$	M1 M1 M1 A1 Cso [4] [M1 M1 M1 A1]
(c)	$\left[\cos 3A = 4\cos^3 A - 3\cos A \Rightarrow 2\cos 3A = 8\cos^3 A - 6\cos A\right]$ $8\cos^3\left(\dfrac{\theta}{2}\right) - 6\cos\left(\dfrac{\theta}{2}\right) - 1 = 0 \Rightarrow 2\cos 3\left(\dfrac{\theta}{2}\right) - 1 = 0$ $\cos 3\left(\dfrac{\theta}{2}\right) = \dfrac{1}{2} \Rightarrow 3\left(\dfrac{\theta}{2}\right) = \dfrac{\pi}{3}, \dfrac{5\pi}{3}, \dfrac{7\pi}{3}, \left(\dfrac{11\pi}{3}\right)$ $\theta = \dfrac{2\pi}{9}, \dfrac{10\pi}{9}, \dfrac{14\pi}{9}$	M1 M1 M1A1 [4]

Type MS · Integrating trig functions using identities

Answer 6 of 9 · Question 11 part d · 4 marks

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(d)	$I = \int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta = \int_0^{\frac{\pi}{6}} \left(\cos 3\theta + 3\cos\theta - \sin 2\theta\right) \mathrm{d}\theta$	M1
	$I = \left[\frac{\sin 3\theta}{3} + 3\sin\theta + \frac{\cos 2\theta}{2}\right]_0^{\frac{\pi}{6}}$	M1
	$= \left[\frac{\sin 3\left(\frac{\pi}{6}\right)}{3} + 3\sin\left(\frac{\pi}{6}\right) + \frac{\cos 2\left(\frac{\pi}{6}\right)}{2}\right] - \left[\frac{\sin 3(0)}{3} + 3\sin(0) + \frac{\cos 2(0)}{2}\right]$	M1
	$= \left[\frac{1}{3} + \frac{3}{2} + \frac{1}{4}\right] - \left[0 - 0 + \frac{1}{2}\right] = \frac{19}{12}$	A1 [4]
		Total 16 marks

Type MS · Integrating trig functions using identities

Answer 6 of 9 · Question 11 part d · 4 marks

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(c)	M1	Substitutes $2\cos 3\left(\frac{\theta}{2}\right)$ in place of $8\cos^3\left(\frac{\theta}{2}\right)-6\cos\left(\frac{\theta}{2}\right)$ correctly to obtain at least $k\cos 3\left(\frac{\theta}{2}\right)-1=0$ where k is a constant
	M1	Rearranges to obtain $\cos 3\left(\frac{\theta}{2}\right)=\frac{1}{k}$ and takes the inverse cosine to find at least any one **correct** value for $3\left(\frac{\theta}{2}\right)$
	M1	Finds at least one correct value for θ
	A1	Finds all three values with no extra values within range. Ignore any values given outside of the range.
(d)	M1	For a **correct** substitution as shown. $4\cos^3\theta=\cos 3\theta+3\cos\theta$
	M1	For an acceptable attempt to integrate either $\cos 3\theta\rightarrow\pm\frac{\sin 3\theta}{3}$ or $-\sin 2\theta\rightarrow\pm\frac{\cos 2\theta}{2}$
	M1	For substituting both given values the correct way around. This mark can be awarded for a complete substitution SEEN into any changed expression. Award this mark for **correct** integration **and** a value of $\frac{19}{12}$ seen without explicit substitution.
	A1	For $\frac{19}{12}$

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Answer 6 of 9 · Question 11 part d · 4 marks

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Answer 7 of 9 · Question 9 part b · 4 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

number		
9 (a)	$\cos 2\theta = \{\cos\theta\cos\theta - \sin\theta\sin\theta\} = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$ $= 2\cos^2\theta - 1$ *	M1 A1*cso [2]
(b)	$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \cos 2\theta \{\mathrm{d}\theta\}$ $\left[\frac{\sin 2\theta}{2}\right]_{\frac{\pi}{3}}^{\frac{3\pi}{4}} = \frac{\sin 2\left(\frac{3\pi}{4}\right)}{2} - \frac{\sin 2\left(\frac{\pi}{3}\right)}{2} = -\frac{2+\sqrt{3}}{4}$	M1 M1M1A1 [4]
(c)	$2\cos^2\theta - 1 = -\cos\theta$ $2\cos^2\theta + \cos\theta - 1 = 0 \Rightarrow (2\cos\theta - 1)(\cos\theta + 1) = 0 \Rightarrow \cos\theta = \ldots$ $\theta = \frac{\pi}{3}$ $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} -\cos\theta \, \mathrm{d}\theta$ $= \left[-\sin\theta\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$ $= -\sin\frac{\pi}{2} - \left(-\sin\frac{\pi}{3}\right) \left\{= \frac{-2+\sqrt{3}}{2}\right\}$ $\frac{2+\sqrt{3}}{4} - \frac{2-\sqrt{3}}{2}$ $\frac{-2+3\sqrt{3}}{4}$	M1 M1 A1 M1 M1 M1 ddM1 A1 [8]
ALT for last 5 marks	$\int_{-\frac{\pi}{3}+\pi}^{\frac{\pi}{2}} (-\cos\theta - \cos 2\theta)\{d\theta\} \{+\} \int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} (-\cos 2\theta)\{d\theta\}$ $= \left[-\sin\theta - \frac{\sin 2\theta}{2}\right]_{-\frac{\pi}{3}+\pi}^{\frac{\pi}{2}} \{+\} \left[-\frac{\sin 2\theta}{2}\right]_{\frac{\pi}{2}}^{\frac{3\pi}{4}}$ $= \left(-\sin\frac{\pi}{2} - \frac{\sin 2\times\frac{\pi}{2}}{2}\right) - \left(-\sin\frac{\pi}{3} - \frac{\sin 2\times\frac{\pi}{3}}{2}\right) \{+\} \left(-\frac{\sin 2\times\frac{3\pi}{4}}{2}\right) - \left(-\frac{\sin 2\times\frac{\pi}{2}}{2}\right)$ $= \frac{-4+3\sqrt{3}}{4} + \frac{1}{2}$ $= \frac{-2+3\sqrt{3}}{4}$	M1 M1 M1 ddM1 A1
		Total 14 marks

Type MS · Integrating trig functions using identities

Answer 7 of 9 · Question 9 part b · 4 marks

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Part	Mark	Additional Guidance
(a)	M1	For $\cos^2\theta-\sin^2\theta=\cos^2\theta-\left(1-\cos^2\theta\right)$ or $\cos^2\theta-\sin^2\theta=\cos^2\theta+\cos^2\theta-1$
	A1*cso	Obtains the correct result with no errors seen. Must have $\cos 2\theta$ on the lhs in their working, does not have to be in the final line provided there is a chain of reasoning.
(b)	M1	For replacement of $2\cos^2\theta-1$ with $\cos 2\theta$,ignore limits
	M1	For an attempt to integrate to an expression of the form $\pm\frac{\sin 2\theta}{2}$
	M1	Correct substitution of the **given** limits to a **changed** expression in terms of $\sin 2\theta$ and subtracts the correct way around. A correct value from a correctly integrated expression scores this mark, no need to see explicit substitution
	A1	Correct answer. isw
(c)	M1	Correctly equates the two curves.
	M1	Rearranges their 3TQ = 0 and attempts to solve (see general guidance). Can be implied by their correct $\cos\theta$ values, $\frac{1}{2}$ **and** –1 from a correct 3TQ, if their 3TQ is not correct, a clear method must be shown.
	A1	Correct angle found. Ignore extra angles.
	M1	**Main**: For stating the correct integral with "correct" limits of their $\frac{\pi}{3}$ and $\frac{\pi}{2}$ $\pm\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}-\cos\theta\,\mathrm{d}\theta$ and $\pm\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}}(2\cos^2\theta-1)\mathrm{d}\theta$ (the later integral can be implied by later work) (their $\frac{\pi}{3}$ must be positive and < $\frac{\pi}{2}$) **ALT**: Correct integrals $\pm\int_{\frac{\pi}{3}}^{\frac{\pi}{2}}(-\cos\theta-\cos 2\theta)\{\mathrm{d}\theta\}$ and $\pm\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}}(-\cos 2\theta)\{\mathrm{d}\theta\}$ oe
	M1 **A1 on Epen**	For a correct integration of **all** their integrals in $\cos 2\theta$ and/or $\cos\theta$ only. Ignore limits
	M1	For correct **explicit** substitution of their limits into **either changed** expression, subtracts the correct way around – substitution of 0 must be seen unless substitution into their changed function gives 0 Can be implied by $\pm\left(\frac{-2+3\sqrt{3}}{4}\right)$ from correct working
	ddM1	Correct strategy finding the required area: **Main**: For subtracting the modulus of their 2 integrals. oe. Either way around. **ALT**: Add the values of two positive area "$\frac{-4+3\sqrt{3}}{4}$" and "$\frac{1}{2}$" oe Dependent on previous **three** method marks
	A1	Correct answer, simplest form and exact area of R

Type MS · Integrating trig functions using identities

Answer 7 of 9 · Question 9 part b · 4 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Integrating trig functions using identities

Answer 8 of 9 · Question 10 part d · 6 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 4 · find part d in this mark scheme extract

Question	Scheme	Marks
10(a)	$\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)=2\sin A\sin B$ *	M1A1cso [2]
(b)	$[A-B=5\theta,\ A+B=9\theta\Rightarrow A=7\theta,\ B=2\theta]$ $\cos 5\theta-\cos 9\theta=2\sin 7\theta\sin 2\theta$ *	B1 cso [1]
(c)	$\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta\Rightarrow 0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$ $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1A1 M1 M1A1 M1A1 [7]
(d)	$\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$ $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,\mathrm{d}x=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,\mathrm{d}x\right]$ $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,\mathrm{d}x=4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ $4\left[\frac{\sin 5x}{5}-\frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}=4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5}-\frac{\sin 9\times\frac{\pi}{7}}{9}\right)-\left(\frac{\sin 0}{5}-\frac{\sin 0}{9}\right)\right]$# $=0.9729...\approx 0.973$	M1 M1 M1M1 M1A1 [6]
		Total 16 marks

Type MS · Integrating trig functions using identities

Answer 8 of 9 · Question 10 part d · 6 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 4 · find part d in this mark scheme extract

Question	Notes	Marks
10(a)	From the formula sheet $\cos(A-B)=\cos A\cos B+\sin A\sin B$ $\cos(A+B)=\cos A\cos B-\sin A\sin B$ Subtracts the two equations to give: $\cos(A-B)-\cos(A+B)=\sin A\sin B-(-\sin A\sin B)$	M1
	For the correct identity as shown with no errors, $\cos(A-B)-\cos(A+B)=2\sin A\sin B$ *	A1 cso [2]
(b)	For finding the value of A and the value of B $A-B=5\theta,\quad A+B=9\theta$ $\Rightarrow A=7\theta,\ B=2\theta$ Or as a minimum: $\cos(7\theta-2\theta)-\cos(7\theta+2\theta)=2\sin 7\theta\sin 2\theta$ *	B1cso [1]
(c)	Sets $\sqrt{3}\sin 7\theta=2\sin 7\theta\sin 2\theta$	M1
	Achieves the correct equation allow the terms in any order. $0=2\sin 7\theta\sin 2\theta-\sqrt{3}\sin 7\theta$	A1
	Factorises their equation $0=\sin 7\theta\left(2\sin 2\theta-\sqrt{3}\right)\Rightarrow \sin 7\theta=0,\ 2\sin 2\theta-\sqrt{3}=0$	M1
	For finding at least **one** correct value for θ using $\sin 7\theta=0$ $\sin 7\theta=0\Rightarrow 7\theta=0,\ \pi,\ 2\pi\Rightarrow\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	M1
	For both correct values $\theta=\frac{\pi}{7},\ \frac{2\pi}{7}$	A1
	For finding **one** correct value of θ using $2\sin 2\theta-\sqrt{3}=0$ $2\sin 2\theta-\sqrt{3}=0\Rightarrow\sin 2\theta=\frac{\sqrt{3}}{2}\Rightarrow 2\theta=\frac{\pi}{3},\ \frac{2\pi}{3}\Rightarrow\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	M1
	For both correct values $\theta=\frac{\pi}{6},\ \frac{\pi}{3}$	A1 [7]
(d)	Uses the identity for $\tan 2x=\frac{\sin 2x}{\cos 2x}\Rightarrow\tan 2x\cos 2x=\sin 2x$	M1
	Substitutes the above into $8\sin 7x\cos 2x\tan 2x$ to give $\int_0^{\frac{\pi}{7}}8\sin 7x\sin 2x\,\mathrm{d}x=\left[\int_0^{\frac{\pi}{7}}4\times(2\sin 7x\sin 2x)\,\mathrm{d}x\right]$ Ignore integral sign and limits for this mark.	M1
	For substituting $\cos 5x-\cos 9x$ for $2\sin 7x\sin 2x$ to give $\int_0^{\frac{\pi}{7}}4(\cos 5x-\cos 9x)\,\mathrm{d}x$ Ignore integral sign and limits for this mark.	M1

Type MS · Integrating trig functions using identities

Answer 8 of 9 · Question 10 part d · 6 marks

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	Integrates $\int_0^{\frac{\pi}{7}} 4(\cos 5x - \cos 9x)\,dx = 4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}}$ Ignore limits for this mark. As a minimum they must obtain: $\left(\pm\frac{\sin 5x}{5} \pm \frac{\sin 9x}{9}\right)$ for the integration.	M1
	Substitutes the limits the correct way around $4\left[\frac{\sin 5x}{5} - \frac{\sin 9x}{9}\right]_0^{\frac{\pi}{7}} = 4\left[\left(\frac{\sin 5\times\frac{\pi}{7}}{5} - \frac{\sin 9\times\frac{\pi}{7}}{9}\right) - \left(\frac{\sin 0}{5} - \frac{\sin 0}{9}\right)\right]$ $= (0.9729...)$	M1
	For the correct value of $\int_0^{\frac{\pi}{7}} 8\sin 7x\cos 2x\tan 2x\,dx = 0.973$	A1 [6]
		Total 16 marks

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Answer 8 of 9 · Question 10 part d · 6 marks

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Answer 9 of 9 · Question 10 part d · 4 marks

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Question	Scheme	Marks
10(a)(i) (ii)	$\sin(A+A)=\sin A\cos A+\cos A\sin A=2\sin A\cos A$ *	B1 cso
	$\cos(A+A)=\cos A\cos A-\sin A\sin A$	
	$\Rightarrow \cos 2A=\cos^2 A-\sin^2 A$	M1
	$=\cos^2 A-(1-\cos^2 A)$	A1*cso
	$=2\cos^2 A-1$ *	[3]
(b)	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{1}{\cos^2\theta}-\frac{\cos^2\theta}{\cos^2\theta}}$ or $\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}\times\frac{\cos^2\theta}{\cos^2\theta}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$	dM1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)$ or $\left(\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
ALT	$\left(\frac{2\tan\theta}{1+\tan^2\theta}=\right)\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$	M1
	$=\frac{2\frac{\sin\theta}{\cos\theta}}{1+\tan^2\theta}$	dM1
	$\cos^2\theta+\sin^2\theta=1\Rightarrow\tan^2\theta+1=\frac{1}{\cos^2\theta}\Rightarrow\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}\left(=2\frac{\sin\theta}{\cos\theta}\times\cos^2\theta\right)=2\sin\theta\cos\theta$	ddM1
	$=\sin 2\theta$ *	A1*cso [4]
(c)	$\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)}=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)\Rightarrow\frac{5}{2}\sin\left(2\left(x+\frac{\pi}{6}\right)\right)=\left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$	M1
	$1-2\cos^2\left(x+\frac{\pi}{6}\right)=-\cos\left(2x+\frac{\pi}{3}\right)$	M1
	$\left(\frac{5}{2}\sin\left(2x+\frac{\pi}{3}\right)=-\cos\left(2x+\frac{\pi}{3}\right)\Rightarrow\right)\tan\left(2x+\frac{\pi}{3}\right)=-\frac{2}{5}$	M1
	$\left(\Rightarrow 2x+\frac{\pi}{3}=\right)-0.38050,\quad 2.76108$	M1
	$(\Rightarrow x=)-0.714,\quad 0.857$	A1A1 [6]

Type MS · Integrating trig functions using identities

Answer 9 of 9 · Question 10 part d · 4 marks

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(d)	$I = \int_0^{\frac{\pi}{2}} (2\sin 2\theta - \cos 5\theta + 2)\ \{\mathrm{d}\theta\}$	M1
	$= \left[\frac{-2\cos 2\theta}{2} - \frac{\sin 5\theta}{5} + 2\theta\right]_0^{\frac{\pi}{2}}$	M1
	$= \left(-\cos 2\left(\frac{\pi}{2}\right) - \frac{\sin 5\left(\frac{\pi}{2}\right)}{5} + 2\times\left(\frac{\pi}{2}\right)\right) - \left(-\cos 2(0) - \frac{\sin 5(0)}{5} + 2\times 0\right)$	M1
	$= \left(1 - \frac{1}{5} + \pi\right) - (-1 - 0 + 0) = \frac{9}{5} + \pi$	A1 [4]
		Total 17 marks

Type MS · Integrating trig functions using identities

Answer 9 of 9 · Question 10 part d · 4 marks

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Part	Mark	Notes
(a)	B1 cso	For $\sin(A+A)$ or $\sin 2A = \sin A\cos A + \cos A\sin A = 2\sin A\cos A$ * No steps missing or incorrect work. Condone $\sin A + A$ with brackets missing if all other work correct.
	M1	For $\cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A$
	A1* cso	For $\cos(A+A)$ or $\cos 2A = \cos A\cos A - \sin A\sin A \Rightarrow \cos^2 A - \sin^2 A = \cos^2 A - (1-\cos^2 A)$ $= 2\cos^2 A - 1$ No steps missing, no incorrect work. Condone $\cos A + A$ with brackets missing if all other work correct.
Students who miss 'A' in their working, (eg coscos – sinsin) may be awarded M mark, but not A or B marks.		
(b)	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity. Allow the 2 to be missing for this mark **but not the final A mark.**
	dM1	For $1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{\cos^2\theta + \sin^2\theta}{\cos^2\theta}$ or multiplying by $\frac{\cos^2\theta}{\cos^2\theta}$ to give $\left(\frac{2\frac{\sin\theta}{\cos\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}} = \right) \frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta}$ The $\times\frac{\cos^2\theta}{\cos^2\theta}$ does not need to be explicitly written. Dependent on previous method mark.
	ddM1	For minimum steps (and dependent on previous method marks): $\frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}}$ or $\frac{2\sin\theta\cos\theta}{\cos^2\theta+\sin^2\theta} = 2\sin\theta\cos\theta$
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
ALT	M1	Uses $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta}$ in the given identity.
	dM1	For replacing $1 + \frac{\sin^2\theta}{\cos^2\theta}$ with $1 + \tan^2\theta$ Dependent on previous method mark.
	ddM1	For replacing $\tan^2\theta + 1 = \frac{1}{\cos^2\theta} \Rightarrow \frac{2\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos^2\theta}} = 2\sin\theta\cos\theta$ Dependent on both previous method marks.
	A1*cso	Achieves the required result – minimum steps as shown, no errors. **Steps in brackets do not need to be shown.**
Students who miss θ in their working, may be awarded M marks, but not the A mark.		

Type MS · Integrating trig functions using identities

Answer 9 of 9 · Question 10 part d · 4 marks

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Allow work throughout where candidates have let, for example $\theta = x + \frac{\pi}{6}$ and worked in θ and 2θ throughout in (c)		
(c)	M1	Rearranges the given equation into $\frac{5\tan\left(x+\frac{\pi}{6}\right)}{1+\tan^2\left(x+\frac{\pi}{6}\right)} = \left(1-2\cos^2\left(x+\frac{\pi}{6}\right)\right)$ **and** changes the left hand side into $P\sin\left(2\left(x+\frac{\pi}{6}\right)\right)$ where P is any constant.
	M1	Correctly states or uses $1-2\cos^2\left(x+\frac{\pi}{6}\right) = -\cos\left(2x+\frac{\pi}{3}\right)$ (condone being replaced with $-\cos 2A$)
	M1	For $\tan\left(2x+\frac{\pi}{3}\right) = Q$ where Q is any constant.
	M1	For -0.38050 or 2.76108 This may be awarded for either value being seen, ignore any extra values seen. Accept any values with at least 3 significant figures.
	A1	For $(x=)-0.714$ or $(x=)0.857$ This may be awarded for either value being seen, ignore any extra values seen. Accept any value with more than 3 significant rounding to those given.
	A1	For $(x=)-0.714$ and $(x=)0.857$ Ignore any extra values **out of range**. Extra values in range will mean A0. Accept any value with more than 3 significant rounding to those given.
Students must use θ or $x+\frac{\pi}{6}$ and 2θ or $2\left(x+\frac{\pi}{6}\right)$		
(d)	M1	For changing the integral into $(I=)\int_0^{\frac{\pi}{2}}(2\sin 2\theta - \cos 5\theta + 2)\,\{\mathrm{d}\theta\}$ $\mathrm{d}\theta$ may be omitted. If the integral sign and limits are not present, these can be implied by a clear attempt in later work to integrate all the terms and a substitution of limits. Students must use 2θ and 5θ in their working.
	M1	For a minimum of 2 out of the 3 terms shown above correctly integrated. Students must use 2θ and 5θ in their working.
	M1	For substituting in the given limits the correct way around into their changed function with at least two terms, there must be at least one explicit substitution of each limit. If the candidate has a constant term as part of the function, they can omit this from this step (as the terms cancel out anyway). This mark may be implied by a correct answer. If the candidate is working with an incorrect function and substitution of 0 clearly gives 0, this substitution does not need to be shown.
	A1	For $\frac{9}{5}+\pi$ The approximate decimal value 4.941…will **usually** indicate correct working (as a guide), but a substitution of limits will be needed to get the 3rd method mark.
In this question, students are directed to use calculus, so must show a correct integration. The penultimate method mark may be implied from a fully correct answer, but if the answer is not correct, we must see substitution of limits as stated.		

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Question	Scheme	Marks
	As with many log questions, there are a number of ways of reaching a correct answer. **The first scheme shows 3 common routes, with a set of general principles in the notes.** **Not all variations in methods will have been captured, please mark to general principles, referring ANY work where this is not possible and worthy of marks to REVIEW.** **This includes work that converts everything to completely new bases.**	